

Lab 02 - First Verilog

In this lab, you've learned how to do an initial and simple design in Verilog to learn the Vivado tooling and process involved in RTL/FPGA design.

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Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

In this lab we got a general understanding of how to use Vivado by loading in a pregenerated file, and writing a code to turn on an LED. We then loaded the program onto the Basys3 circuit board and got to witness the LED correctly turn on.

Lab Questions

1 - Describe the stages of building a Verilog project in Vivado.

The first stage was to set up the project and create the different files we would be using in this project. We used the pre-existing top and test code to simulate and run the program. Once the code was created and the hardware that we would be using was connected. We then simulated, and synthesized the project before creating a bitstream and loading it into the circuit board.

2 - What is the value in looking at the elaborated design schematic?

The value of looking at the elaborated design schematic is seeing where logic failures might occur.

3 - Why should we simulate our designs frequently? What does the simulation do?

It assists us in debugging prior to uploading the code to the hardware which could be time extensive, and helps us understand the expected outcome of our code prior to running.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.

https://github.com/MatthewGrundyBSU/ECE_Week2